

AI Course

Capstone Project Action Plan

For students (instructor's review required)

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Course	AI Course
Team Name	Cogni
Team Leader/ Members	Marwan Elamami (Team Leader) / Abdulrahman Khengari, Ahmed Alaeb, Sanad Ali, Mohammed Thabet, Abdul Majid
Project Title	Bayan: An AI Reading Assistant for Arabic Readers with Dyslexia
Goal	• Help Arabic readers with dyslexia read and understand Arabic text on their own. • Simplify complex Arabic text automatically without changing its meaning. • Add full diacritics (tashkeel) so every word can be pronounced correctly. • Read the text aloud with synchronized word-by-word highlighting. • Keep the core simplification model(s) small enough to run on-device.
Abstract	Arabic creates specific barriers for readers with dyslexia: short vowels are usually omitted, prefixes and suffixes attach directly to words, and long sentences chain many clauses together. Most dyslexia tools target Latin-script languages; we found no deployed Arabic dyslexia-support software. Bayan is a three-stage AI pipeline that simplifies complex Arabic text with a fine-tuned Arabic encoder-decoder Transformer, restores its diacritics, and reads it aloud with synchronized word highlighting. Because no large open Arabic simplification corpus exists, we build our own from the BAREC readability corpus, keeping only rewrites that automatic validators and human reviewers confirm are faithful and easier to read.
Method	1. Data construction: easy BAREC sentences and religious scripture are kept unchanged. For each hard sentence, an LLM generates several candidate simplifications; an LLM judge calibrated on human annotations scores meaning preservation, and a fine-tuned Arabic BERT classifier checks readability. The best passing candidate is kept, and annotators review samples. 2. Simplification: fine-tune one or more pretrained Arabic encoder-decoder Transformers that can be quantized for on-device use. 3. Diacritization: a ready pretrained model, checked on simplified text so base letters are never altered; fine-tuned only if time permits. 4. Read-aloud: a ready pretrained Arabic text-to-speech model with word timestamps for synchronized highlighting. 5. Evaluation: compare against a copy-the-input baseline, a reproduced published system, and large LLMs using SARI, readability reduction (independent BAREC model), meaning preservation, DER/WER, and human evaluation.

6. **Application:** one app connecting all stages, with an optional quality gate that shows the original text when a simplification is judged unfaithful.

Data	Sources • BAREC (CAMEL Lab, CC BY-SA 4.0, public on Hugging Face): 69,441 Arabic sentences with readability labels: the complex sentences to simplify and the labels for the readability classifier. • Bayan simplification corpus (built by the team): complex-to-simple pairs generated with LLM APIs, filtered by the validators, and reviewed by team annotators. A 2% pilot run is already complete. • Tashkeela: fully diacritized Arabic text for validating the diacritization stage. • Evaluation sets: a held-out test set verified by the team, plus external human-written test sets (the SAMER test split, used for evaluation only with CAMEL Lab's approval, and DAASI). How we use it Training the simplification model(s), calibrating the validators and the readability classifier, validating the diacritization stage, and benchmarking Bayan against published baselines.
Expected Outcome	Deliverables • A working prototype: the user enters Arabic text and receives a simplified, fully diacritized version that is read aloud with synchronized word highlighting. • Fine-tuned simplification model(s) small enough for on-device use, benchmarked against published baselines and large language models. • A new Arabic simplification corpus and evaluation benchmark. Benefits • Greater reading independence and comprehension for Arabic readers with dyslexia. • Support for students, teachers, and schools, as well as readers with low literacy and learners of Arabic. • An on-device core model that works offline and keeps users' text private.
Role by Member	Marwan Elamami Team Leader & Application Lead Project management; system architecture and integration; application development; read-aloud with word highlighting. Abdulrahman Khengari Data Lead Building and validating the simplification corpus; data quality control; annotation guidelines. Ahmed Alaeb Model Training Lead Training the simplification model(s) for on-device use; integrating a ready diacritization model (fine-tuning it if time permits). Sanad Ali Machine Learning Engineer Co-training the simplification model(s); data annotation and quality review. Mohammed Thabet Application Engineer Application development; read-aloud with word highlighting; data annotation and quality review.

Abdul Majid | [Evaluation & Benchmarking Lead](#)

Test set design; benchmarking against published baselines; automatic and human evaluation.

Schedule Summary	<i>This schedule is a working draft; tasks and timing may be adjusted as the project progresses.</i> • Week 1 (D1–D5): finalize the data pipeline and validators; write annotation guidelines; design the test sets; set up training and baselines; build the app skeleton and choose the TTS engine; contact dyslexic readers and schools for user testing. • Week 2 (D6–D10): generate and annotate the full corpus; first simplification training run; evaluate ready diacritization models on simplified text; first benchmark results; connect simplification and diacritization to the app; word timings. • Week 3 (D11–D15): improved training runs; optional diacritization fine-tuning if time permits; full benchmark with human evaluation; end-to-end integration with read-aloud and highlighting; optional quality gate. • Week 4 (D16–D20): freeze models; final evaluation; user testing with dyslexic readers; bug fixing; final report and presentation. Detailed tasks, owners, and daily status are tracked in the WBS workbook.
Comment & Assessment	Data: Define the corpus size, pair-selection criteria, annotator agreement, and leakage-safe train/validation/test split before LLM-generated examples are used for training. Evaluation: Run human evaluation with Arabic readers with dyslexia, including comprehension, reading effort, and acceptability, with informed consent and privacy safeguards. Measure diacritization error and base-letter changes on held-out simplified text, and report on-device latency, size, and quality trade-offs. GenAI option: Keep LLM generation behind a human-reviewed quality gate and retain source text and validation evidence for each accepted simplification.